

	Biogeochemistry Laboratory	Date: May 10, 2026	SOP No. PH-001
	Standard Operating Procedure	Title: Calibration of pH Meter	
	Approved By: Dr. Jane Smith	Revised: 2026-05-10	

Contents

1	Purpose	2
2	Scope	2
3	Required Materials	2
4	Procedure	2
4.1	Initial Inspection	2
4.2	Calibration	2
5	Documentation	2
6	References	2

1. Purpose

- 1.1 This SOP describes the procedure for calibrating laboratory pH meters prior to analytical measurements.
- 1.2 Calibration ensures measurement accuracy and consistency between laboratory personnel.

2. Scope

- 2.1 This procedure applies to all benchtop pH meters used within the laboratory.

3. Required Materials

- pH meter
- Calibration buffers (4, 7, and 10)
- Deionized water
- Kimwipes

4. Procedure

4.1. Initial Inspection

- 4.1 Inspect the electrode for cracks, salt buildup, or air bubbles.
- 4.2 Verify the electrode storage solution is filled appropriately.

4.2. Calibration

- 4.3 Rinse the probe using deionized water.
- 4.4 Immerse the electrode into the pH 7 buffer solution.
- 4.5 Wait for the reading to stabilize before accepting calibration.
- 4.6 Repeat the process using pH 4 and pH 10 buffers.

5. Documentation

- 5.1 Record calibration date, operator initials, and slope values in the laboratory logbook.

6. References

- 1. Manufacturer operating manual

2. EPA Method 150.1